SHORTY SHORT
FOOTBALL ANALYTICS

INDEPENDENT NFL RESEARCH / COMPLETE FORECAST

THE BRONCOS & THE FIELD

Six forecasting lenses. A full NFL season.
The uncertainty, tested and explained.

12 million primary component seasons

4.5 million matched stress evaluations

32 teams / 179 remaining regular-season games

Seven executive summaries for the full package

Forecast information frozen: 20 October 2026, 20:37 UTC
1:37 p.m. Pacific Daylight Time

Edition 2 / Consolidated research report
Assembled 21 October 2026. Later market captures are labeled comparators.

Independent analysis. NFL and Broncos marks identify the subjects; no affiliation is implied.

COMPLETE-PACKAGE EXECUTIVE OVERVIEW

A contender, with a difficult path

69.4%	26.9%	8.5%	4.2%
Make the playoffs	Win the AFC West	Win the AFC	Win the Super Bowl

Denver projects to 10.27 wins and ranks eighth in championship probability. The most common exact record is 10-7, but it occurs in only 19.2% of the forecast distribution. The 10th and 90th percentiles are 8 and 13 wins. The Broncos are a credible contender; most simulated seasons still end before the Super Bowl.

The league favorite is San Francisco at 13.2%. Kansas City and Buffalo are effectively tied near 9.4%, followed by Minnesota at 8.6% and Jacksonville at 8.2%. The most frequent individual Super Bowl pairing is Kansas City-San Francisco, at 4.74% of all seasons. It is one of many plausible endings.

Denver's leading first playoff opponents are Kansas City and Jacksonville. Conditional on Denver qualifying, they account for 15.2% and 14.5% of first opponents. Conditional on Denver winning the AFC, San Francisco is its leading Super Bowl opponent at 26.7%. Qualification, opponent and matchup-win probabilities answer different questions.

What changed from the initial report

The primary forecast now combines six separately fitted families, including Bayesian team-strength updating and dynamic availability. The positional possession model remains a documented challenger after unfavorable longer-horizon diagnostics. New adversarial tests examine shared market inputs, strength evolution, clustered absences and engine edge cases.

Reading order: seven audience summaries follow; then Denver, the full league, players, markets, model construction, validation, stress tests and reproducibility. All seven summaries address this completed package. Primary forecast evidence: F1-F3; methodology: M1-M7. Information cutoff: 20 October 2026, 20:37 UTC.

REPORT NAVIGATION

Find your view of the forecast

Seven summaries of the completed package

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The evidence behind those summaries

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The PDF also contains bookmarks. The offline HTML has direct section links. M1-M7 identify prediction methods; F1-F9 identify computed evidence; S references identify external primary sources. The same frozen forecast underlies every audience summary.

EXECUTIVE SUMMARY / COACHING STAFF

Preparation priorities and the postseason path

The decision that matters

Denver has a credible playoff path; use the forecast to prioritize preparation questions and contingency planning.

At the October 20 information cutoff, Denver averages 10.27 wins, with an 8-13 win range between the 10th and 90th percentiles. The probabilities are 69.4% to make the playoffs, 26.9% to win the AFC West, 8.5% to win the AFC, and 4.21% to win the Super Bowl. Kansas City is the most frequent first playoff opponent, conditional on qualification; San Francisco leads potential Super Bowl opponents, conditional on Denver reaching that game.

The Week 8 home game against Kansas City is close to even at 51.7%. Within the simulated seasons, division probability is 43.6% when Denver wins that game and 8.9% when it loses. Those conditionals describe associated season paths, not the causal payoff of a specific intervention. League-wide, San Francisco leads title probability at 13.25%, followed by Kansas City, Buffalo, Minnesota, and Jacksonville.

Denver’s adjusted unit proxies highlight strong pass defense and sack avoidance alongside weaker net-passing production. They support film questions about protection, decisions, route execution, and opponent matchups; they do not isolate individual performance. Surtain-Hufanga return scenarios and the Bayesian availability process support contingency comparisons. Return dates are assumptions, and partial ensemble sensitivities change only the availability family. Medical and tactical judgments require information beyond these public-data diagnostics.

The six-family forecast pools 12 million retained component seasons. The positional possession simulator remains experimental because incremental predictive benefit was not established. Market-only performed better on the exposed 2025 near-term diagnostic; excluding both quote-fed families barely changes Denver’s title probability, but materially changes some other teams. Combined absence clustering and delayed Surtain-Hufanga returns lower the one-family partial playoff estimate to 68.95%; this is a contingency sensitivity.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

EXECUTIVE SUMMARY / PLAYERS

A meaningful opportunity across many possible seasons

The decision that matters

The forecast describes opportunity and uncertainty; it does not assign effort, credit, blame, or individual limits.

Denver reaches the playoffs in about 69 of every 100 weighted season outcomes using the October 20 information set. The division chance is 26.9%, the AFC-title chance 8.5%, and the Super Bowl chance 4.21%. The average finish is 10.27 wins. A 10-7 record is the single most common result, but occurs only 19.2% of the time; the 10th-90th percentile span is 8-13 wins.

The path remains open across the league. San Francisco has the highest title probability at 13.25%, with Kansas City and Buffalo near 9.4% each. If Denver qualifies, Kansas City and Jacksonville are the most frequent opening opponents. If Denver wins the AFC, San Francisco and Minnesota are the most frequent Super Bowl opponents. These are conditional possibilities, not a fixed bracket waiting to happen.

The descriptive unit estimates identify pass defense as a strength and offensive passing production as an area to examine. They combine teammates, opponents, decisions, and scheme; no unit rank measures a single player’s ability. Named availability cases consider Pat Surtain II and Talanoa Hufanga returning in Week 7 or remaining absent through Week 10. Another case includes Pat Bryant at earliest eligibility. Those assumptions do not forecast recovery dates, and their partial ensemble effects change only one model family.

The main forecast combines six model families and changing team-strength uncertainty. Injury scenarios explore participation assumptions without predicting recovery dates. The experimental positional simulator receives no ensemble weight because added forecasting value remains unproven. Market forecasts performed better near-term in the previously exposed 2025 check, so precision should not be confused with certainty. One-family stresses leave the partial playoff estimate between 68.95% and 69.42%; these scenarios do not bound every risk to the team.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

EXECUTIVE SUMMARY / REPORTERS

The publishable finding and the evidence behind it

The decision that matters

Denver is an eighth-ranked title contender by modeled probability, with a roughly seven-in-ten playoff chance.

Using information frozen at 20:37 UTC on October 20, 2026, Shorty Short Football Analytics estimates Denver at 69.4% to reach the playoffs, 26.9% to win the AFC West, 8.5% to win the AFC, and 4.21% to win the Super Bowl. Denver averages 10.27 wins and ranks eighth by title probability. Its modal 10-7 record represents just 19.2% of outcomes.

The all-league favorite is San Francisco at 13.25%, followed by Kansas City at 9.43% and Buffalo at 9.41%. The most frequent Super Bowl pairing is Kansas City-San Francisco at 4.74%. Denver's most frequent first playoff opponent is Kansas City, 15.2% conditional on qualification. Its leading Super Bowl opponent is San Francisco, 26.7% conditional on winning the AFC. Those denominators must accompany the figures.

Six families combine market, Elo, public efficiency, Bayesian strength, availability, and hybrid evidence through 12 million retained component seasons. Their exact weighted mixture creates no additional simulations. The positional possession challenger remains outside the forecast because its incremental benefit was not established. Player-specific findings concern documented availability and conditional Surtain-Hufanga return scenarios. Descriptive unit ranks combine team and opponent effects; they cannot identify individual talent or predict a medical return date.

A candid validation finding is that market-only beat the ensemble near-term in the previously exposed 2025 season; longer-horizon differences were unresolved. Neither exposed diagnostics nor game-level calibration establish championship calibration. Removing both quote-fed families and refitting leaves Denver's title chance near 4.19%, while Dallas and the Giants move materially. Post-cutoff futures comparisons establish no live trading edge. Registered one-family dynamic stresses span 68.95-69.42% playoffs and 4.18-4.25% titles; these are scenario sensitivities, not calibrated uncertainty intervals.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

EXECUTIVE SUMMARY / FANS

Denver has a playoff path in a wide-open field

The decision that matters

A playoff run is more likely than a miss, while a championship still requires several difficult steps.

The October 20 forecast gives Denver a 69.4% playoff chance, 26.9% to win the AFC West, 8.5% to take the AFC, and 4.21% to lift the Super Bowl trophy. That is eighth-best in this model. The average finish is 10.27 wins; 10-7 is the most common exact record, but only about one season in five ends there. The 10th-90th percentile range runs from eight to thirteen wins.

The schedule contains real opportunities and close calls: Denver is 63.5% to win at Arizona, 51.7% at home against Kansas City, and 50.7% at home against Buffalo. If Denver makes the playoffs, Kansas City and Jacksonville are its most frequent opening opponents. If it wins the AFC, San Francisco leads potential Super Bowl opponents. Neither list guarantees a matchup.

The rest of the NFL keeps this in perspective. San Francisco leads the championship forecast at 13.25%; Kansas City and Buffalo are around 9.4% each, followed by Minnesota and Jacksonville. Even the favorite loses most simulated title races. Player-specific analysis examines Surtain and Hufanga availability scenarios alongside Denver’s descriptive unit strengths. Those scenarios do not predict recovery dates, and team-level unit measures cannot isolate individual ability.

The main answer comes from six families and 12 million retained simulated seasons, with changing strength and availability uncertainty. The more elaborate positional simulator was tested but did not earn a place in the final blend. Markets did better near-term in the exposed historical check. Excluding both quote-fed families still leaves Denver’s title estimate around 4.2%; that is reassuring for this sensitivity, not proof against every forecasting error. One-family strength and absence stresses keep the partial playoff estimate around 69%, while leaving broader forecasting risks outside that narrow test.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

EXECUTIVE SUMMARY / QUANTS

Joint predictive pooling, chronology, and model risk

The decision that matters

The forecast is a coherent empirical mixture with audited numerical precision; its rare-event calibration remains unestablished.

The frozen six-family mixture puts Denver at 69.37% playoffs, 26.87% division, 8.47% AFC, and 4.208% Super Bowl, with 10.268 expected wins. San Francisco leads title probability at 13.25%; Kansas City-San Francisco is the largest joint Super Bowl pairing at 4.735%. Opponent conditionals are pooled joint numerators divided by pooled denominators, preserving selection effects and avoiding averages of conditional percentages.

Hybrid, Elo, market, efficiency, Bayesian strength, and availability families contribute complete-season laws. Efficiency uses public EPA and success rates. Model identity persists across a season. Reconstructed 2023 predictions train the regularization comparison; 2024 selects lambda, and final weights refit on 2023-24. Lambda 0.1 shrinks toward equal-family weights. The mixture exactly aggregates 12 million retained component worlds; its 0.0141-percentage-point Denver-title Monte Carlo SE bound excludes model, source, parameter-bank, and weight uncertainty.

The exposed 2025 near-term three-class log losses are 0.65157 for the ensemble and 0.63757 for market-only. Multiweek differences are unresolved by target-week clustered intervals. There is no untouched test season, and game scores cannot certify championship calibration. The positional possession challenger failed to establish incremental value and has zero voting weight. Unit features remain team/opponent proxies; player availability cases do not identify causal individual value or clinical return probabilities.

Removing market and hybrid eliminates 39.33% of frozen family mass. Fixed renormalization gives Denver 4.168% title probability; refitting four families at unchanged lambda gives 4.188%. This removes directly quote-fed families without purging inherited research lineage. Clone-label invariance holds when family mass is preserved; highly correlated forecast errors rule out six-independent-evidence claims. Quote timestamps cannot establish staleness or an executable edge. Across 4.5 million scenario-season evaluations, one-family partial title sensitivities span 4.180-4.251%; paired Monte Carlo errors quantify simulation differences only.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

EXECUTIVE SUMMARY / PREDICTION-MARKET PARTICIPANTS

Separate event forecasts, displayed quotes, and executable value

The decision that matters

The retained forecast provides dated event probabilities and sensitivity evidence, without establishing a current tradable edge.

At the October 20, 20:37 UTC cutoff, Denver's model probabilities are 69.37% playoffs, 26.87% AFC West, 8.47% AFC champion, and 4.208% Super Bowl champion. It averages 10.27 wins. San Francisco leads the all-league title field at 13.25%, while the largest Super Bowl pairing, Kansas City-San Francisco, has only 4.74% probability. Event definitions and joint paths matter when interpreting contracts with different settlement conditions.

The retained Kalshi Denver quotes show YES bid-ask ranges of 70-71 cents for playoffs, 29-30 for the division, 9-10 for the conference, and 4-5 for the title. They were observed about eight to nine minutes after the model cutoff. Individual quote age, current fills, full costs, and complete settlement conditions were not verified. A midpoint is descriptive; it is not an executable purchase price. These observations do not establish a current net edge.

The forecast pools six families across 12 million retained complete-season simulations. Market information enters both market and hybrid families. Removing both and refitting the other four barely changes Denver's title estimate to 4.188%, yet moves Dallas playoff probability by -11.21 percentage points and the Giants by +7.86. Market dependence is therefore team-specific. Market-only also beat the ensemble in the exposed 2025 near-term diagnostic; no universal market-beating claim is supported.

Bayesian strength and availability models broaden the forecast beyond fixed ratings, while the unproven positional challenger receives no weight. Player analysis concerns named availability and descriptive team-unit diagnostics; it provides no individual yardage, touchdown, or award-market probabilities. Game-level checks do not validate rare-event championship calibration or eliminate shared model error. Registered one-family dynamic sensitivities leave Denver's partial title estimate at 4.18-4.25%; this range is not a calibrated market-value interval.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

EXECUTIVE SUMMARY / BETTORS

A forecast is useful only with the correct payout and uncertainty

The decision that matters

The retained Denver title price does not clear the model's payout threshold, and no current betting edge is established.

The forecast gives Denver 69.4% playoffs, 26.9% AFC West, 8.5% AFC, and 4.21% Super Bowl, with 10.27 expected wins. San Francisco leads title probability at 13.25%, followed by Kansas City and Buffalo near 9.4%. These probabilities describe the October 20 information set, not current prices or news. A single most-likely finish still leaves substantial variation across season outcomes.

The retained Denver +2000 title quote requires 4.76% to break even under a binary payout without voids or costs. At the model's 4.208%, expected profit is approximately -11.63% of stake before fees or other costs. Denver's 4.03% proportional no-vig share on that sportsbook board is a different quantity; exceeding normalized market share does not mean beating the payout. The observation was after cutoff and is not asserted current.

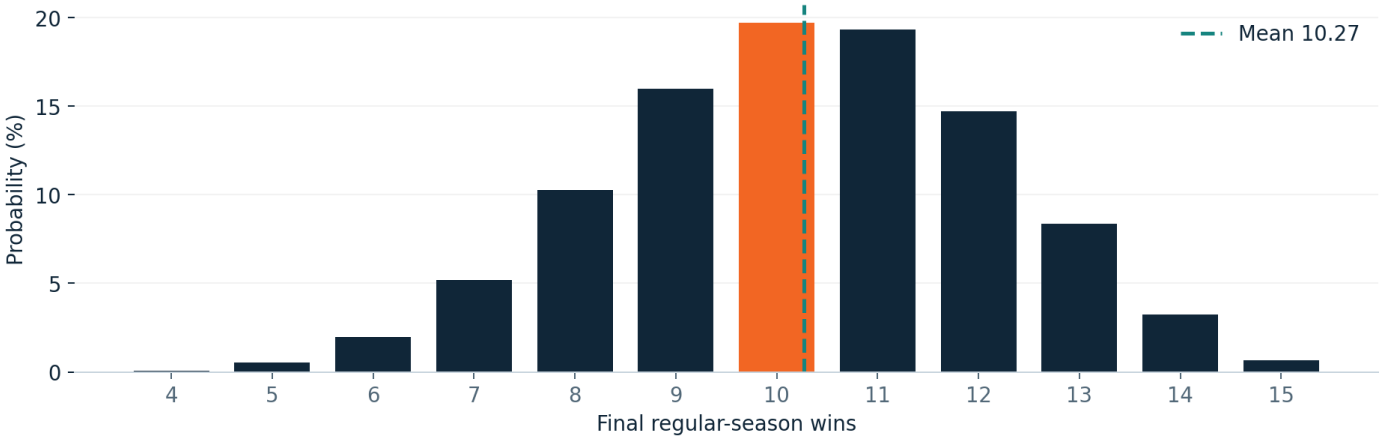
Model risk matters more than tiny Monte Carlo error. Six families contribute 12 million retained simulated seasons. Removing both quote-fed families and refitting gives Denver about 4.19% title probability, but other teams move more. Market-only outperformed the ensemble near-term in exposed 2025; there is no untouched test season or established championship calibration. The experimental positional model did not establish added forecasting value.

Surtain-Hufanga return scenarios change only the availability family; their partial ensemble effects are not full-model causal injury estimates or medical return predictions. Unit ranks describe team/opponent proxies and cannot supply individual prop probabilities. Any future comparison requires matched settlement rules, executable prices, and costs; spread or total numbers alone do not imply a -110 price. The one-family dynamic title sensitivities span 4.18-4.25%, below the recorded 4.76% payout threshold; broader model uncertainty is not quantified by that range.

Scope: the full completed forecast, including validation and adversarial review. All percentages are frozen model estimates. Detailed tables, denominators and methods appear in the corresponding report sections. Evidence: F1-F9.

THE BRONCOS / PRIMARY ENSEMBLE

Denver: the season distribution



Pooled probabilities of final win totals. The dashed line is the mean; the orange bar is the modal win count. F1.

Season event	Probability	Interpretation
Make playoffs	69.4%	Includes division and wildcard qualification
Win AFC West	26.9%	Kansas City remains the division favorite
AFC first seed	8.5%	Potential bye and conference home-field advantage
Reach divisional round	36.8%	Includes a first-seed bye
Reach AFC title game	17.5%	One game away from the Super Bowl
Win AFC / reach Super Bowl	8.5%	Distinct from winning the Super Bowl
Win Super Bowl	4.21%	One champion per simulated season

Method M1: retain 93 completed league scores and simulate all remaining games, standings and playoffs within each family; pool whole-season counters with frozen learned weights. Mean wins are expectations, not a literal record. Discrete 8-13 endpoints can contain more than 80% of the probability. [F1, F2]

THE BRONCOS / REMAINING REGULAR SEASON

Eleven games still to decide it

Week	Matchup	DEN win	Mean DEN	Mean opp.
7	at Arizona Cardinals	63.5%	24.7	20.5
8	vs. Kansas City Chiefs	51.7%	21.5	21.0
9	at Carolina Panthers	47.6%	21.5	22.3
11	vs. Las Vegas Raiders	72.2%	25.6	18.2
12	at Pittsburgh Steelers	45.4%	20.0	21.4
13	vs. Miami Dolphins	76.1%	25.3	16.3
14	at New York Jets	54.8%	21.6	20.0
15	at Las Vegas Raiders	55.6%	22.8	21.0
16	vs. Buffalo Bills	50.7%	24.4	24.1
17	at New England Patriots	40.6%	20.1	23.2
18	vs. Los Angeles Chargers	68.6%	23.8	17.4

Week 10 is Denver's bye. New England away is its hardest remaining game in this forecast, at 40.6% to win. Pittsburgh away is 45.4%; Carolina away is 47.6%. Miami at home is the most favorable, at 76.1%. Kansas City and Buffalo at home are close to even.

The Kansas City game changes the path

In the joint forecast, Denver wins the division in 43.6% of seasons in which it beats Kansas City in Week 8, compared with 8.9% when it loses. This is an association across season paths: those groups also differ in inferred latent strength and later outcomes. It is not an isolated causal value of one coaching decision.

Method M1: each displayed game probability integrates over that game's place in shared season trajectories. Ties are retained separately in the full CSV; win and loss need not sum to 100%. Mean scores are expectations from a statistical score model, not exact-score picks or a scoring-event simulation. [F1]

THE BRONCOS / POSTSEASON OPPONENTS

Who is most likely to stand in the way?

First playoff opponent, given Denver qualifies

Opponent	Share of DEN playoff paths	DEN win if matched
Kansas City Chiefs	15.2%	49.1%
Jacksonville Jaguars	14.5%	41.6%
New England Patriots	11.8%	49.1%
Buffalo Bills	11.4%	46.2%
Pittsburgh Steelers	9.3%	51.0%
Baltimore Ravens	9.2%	48.6%

Super Bowl opponent, given Denver wins the AFC

Opponent	Share of DEN Super Bowls	DEN win if matched
San Francisco 49ers	26.7%	48.1%
Minnesota Vikings	17.9%	50.1%
Seattle Seahawks	10.2%	45.5%
Chicago Bears	7.7%	49.0%
Detroit Lions	7.6%	49.7%
Carolina Panthers	6.3%	53.7%

Denver wins the AFC in 12.2% of its playoff seasons. Once it wins the AFC, it wins the Super Bowl in 49.7% of those seasons. The difficulty is surviving the route there; a low unconditional title probability does not imply a hopeless single Super Bowl matchup.

Method M2: count joint opponent-and-stage events within complete seeded brackets. Pool numerators and denominators before division. First opponent includes a divisional-round opener after a bye. Win-if-matched conditions on both teams surviving; it is not a neutral-site game played today. Displayed top six lists omit other possible opponents. [F1]

THE LEAGUE / PARITY VIEW

AFC: every team, every major outcome

Team / div.	Mean wins	Modal record	Playoffs	Division	Conf.	SB
BUF / East	11.70	12-5	89.2%	60.9%	18.0%	9.4%
NE / East	10.37	10-7	76.4%	36.7%	10.7%	5.5%
NYJ / East	6.61	6-11	10.7%	2.1%	0.6%	0.3%
MIA / East	4.00	4-13	1.0%	0.3%	<0.1%	<0.1%
BAL / North	9.92	10-7	58.3%	29.0%	7.4%	3.8%
PIT / North	9.85	10-7	58.1%	28.4%	6.1%	3.0%
CIN / North	9.73	10-7	55.4%	26.0%	6.1%	3.0%
CLE / North	8.98	9-8	40.4%	16.6%	2.4%	1.0%
JAX / South	10.97	11-6	84.0%	72.7%	15.5%	8.2%
HOU / South	8.41	9-8	35.5%	19.4%	4.0%	2.1%
IND / South	7.06	7-10	12.3%	7.0%	1.0%	0.5%
TEN / South	4.63	4-13	1.6%	0.8%	<0.1%	<0.1%
KC / West	11.79	12-5	90.8%	69.6%	18.4%	9.4%
DEN / West	10.27	10-7	69.4%	26.9%	8.5%	4.2%
LV / West	7.51	7-10	16.2%	3.4%	0.9%	0.4%
LAC / West	4.29	4-13	0.8%	<0.1%	<0.1%	<0.1%

Kansas City leads the West, Buffalo the East and Jacksonville the South. The North is crowded: Baltimore, Pittsburgh and Cincinnati are all near 26-29% to win it. A narrow lead in a marginal division probability is not a confident division pick.

Method M1: all 32 clubs share the same simulated league, so each world has eight division champions, fourteen playoff qualifiers, two conference champions and one title winner. Rows are sorted by expected wins within divisions. Modal records are separate marginal modes and need not jointly form one possible league table. LA means Rams; LAC means Chargers. Positive probabilities below 0.1% remain positive. [F1]

THE LEAGUE / PARITY VIEW

NFC: every team, every major outcome

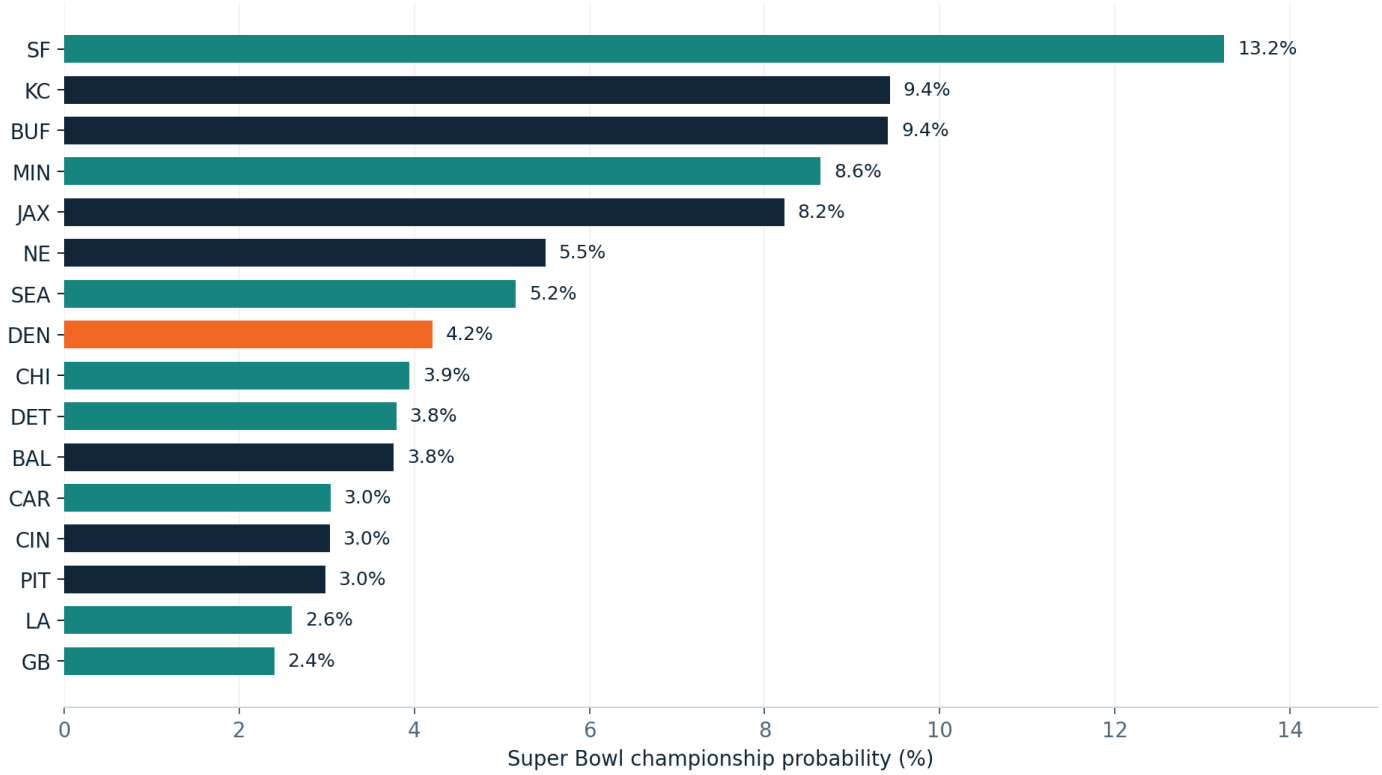
Team / div.	Mean wins	Modal record	Playoffs	Division	Conf.	SB
DAL / East	7.68	8-9	37.6%	34.6%	4.0%	1.9%
NYG / East	7.28	7-9-1	36.7%	33.1%	2.2%	0.9%
PHI / East	6.86	7-10	22.8%	20.9%	1.7%	0.7%
WAS / East	5.74	6-10-1	13.5%	11.4%	0.9%	0.4%
MIN / North	11.91	12-5	91.3%	58.5%	17.8%	8.6%
CHI / North	9.96	10-7	64.9%	14.4%	8.0%	3.9%
DET / North	9.71	10-7	60.9%	17.3%	7.8%	3.8%
GB / North	9.34	9-8	53.7%	9.8%	5.1%	2.4%
CAR / South	9.08	9-8	64.1%	58.9%	6.8%	3.0%
ATL / South	7.13	7-10	24.9%	17.9%	2.4%	1.1%
NO / South	6.76	7-10	18.1%	14.4%	1.3%	0.6%
TB / South	5.80	6-11	11.0%	8.7%	1.0%	0.4%
SF / West	12.79	13-4	96.7%	83.4%	26.1%	13.2%
SEA / West	9.71	10-7	62.4%	12.2%	9.7%	5.2%
LA / West	8.33	9-8	35.4%	4.0%	4.9%	2.6%
ARI / West	6.40	6-11	6.1%	0.5%	0.3%	0.1%

San Francisco has the strongest division and conference position. Minnesota leads a competitive North. Carolina leads the South despite a lower overall title probability than several wildcard contenders. Dallas and the Giants are close in the East; division structure materially affects qualification chances.

Method M1: all 32 clubs share the same simulated league, so each world has eight division champions, fourteen playoff qualifiers, two conference champions and one title winner. Rows are sorted by expected wins within divisions. Modal records are separate marginal modes and need not jointly form one possible league table. LA means Rams; LAC means Chargers. Positive probabilities below 0.1% remain positive. [F1]

THE LEAGUE / TITLE LEADERS

The championship field is broad



Top 16 championship probabilities; every other team remains in the full forecast. AFC: navy. NFC: teal. Denver: orange. [F1]

San Francisco is the leading single team at 13.2%, meaning the model still gives the rest of the league 86.8% combined. Kansas City and Buffalo differ by only 0.02 percentage points; treating that ordering as a decisive football judgment would overstate the evidence.

Denver's eighth-place title ranking is a distributional result, not a claim that it would defeat all lower-ranked teams on a neutral field. Current record, route to a division title, bye chances, home fields, opponent paths and future strength uncertainty all contribute.

Method M1: count the final winner in each completed postseason, then form the weighted mixture. All 32 probabilities sum to 100%. The observed component range and stress cases measure model disagreement; neither is a calibrated confidence interval. [F1-F2]

THE LEAGUE / LIKELY POSTSEASON STRUCTURE

Division favorites and Super Bowl pairings

Division	Marginal favorite	Win division
AFC East	Buffalo Bills	60.9%
AFC North	Baltimore Ravens	29.0%
AFC South	Jacksonville Jaguars	72.7%
AFC West	Kansas City Chiefs	69.6%
NFC East	Dallas Cowboys	34.6%
NFC North	Minnesota Vikings	58.5%
NFC South	Carolina Panthers	58.9%
NFC West	San Francisco 49ers	83.4%

AFC team	NFC team	Pairing probability
KC	SF	4.74%
BUF	SF	4.66%
JAX	SF	4.06%
KC	MIN	3.28%
BUF	MIN	3.18%
NE	SF	2.87%

Method M2: pairing probabilities are unconditional joint frequencies across whole playoffs, not products of independent conference probabilities. A list of marginal favorites is not a most-likely full bracket: retained primary counters do not identify the most frequent complete bracket. Choosing all individual favorites would hide the many alternative paths. [F1]

MARKET COMPARISON / EXPLICITLY TIMED

What the recorded markets say

Denver event	Ensemble	Kalshi YES bid / ask
Playoffs	69.37%	70c / 71c
AFC West	26.87%	29c / 30c
AFC champion	8.47%	9c / 10c
Super Bowl	4.21%	4c / 5c

These futures observations are later comparators. They were captured around 1:44-1:46 p.m. PDT, seven to nine minutes after the 1:37 p.m. forecast cutoff. None entered fitting. Bid/ask ranges are prices, not confidence intervals, and exchange midpoints are not automatically calibrated probabilities.

A no-vig share is not a purchase price

Denver's recorded BetMGM Super Bowl quote was +2000. Its raw break-even probability is 1/21 = 4.76%. Proportional normalization across all 32 listed outcomes gives a 4.03% no-vig share. The model's 4.21% exceeds that normalized share but remains below the probability needed to break even at +2000.

At the model probability, cost-free fair odds would be approximately +2276. A binary +2000 wager has modeled expected net return 21p - 1 = -11.6% per unit staked. Buying a $1 YES payout at 5c has modeled expected value p - 0.05 = -0.79c before costs. These calculations use frozen assumptions and observed prices; they establish no current executable edge.

Method M3: convert American odds to raw break-even probabilities; normalize mutually exclusive sportsbook outcomes only under the stated proportional convention. Compare exchange contracts at their observed purchase side. Fees, full settlement details, current order-book depth and fills were not verified. Source snapshots and exact timestamps: F6; S13-S15. All 160 all-team futures rows are retained.

MARKET COMPARISON / INFORMATION AND EXECUTION

A market forecast is a model, too

The market-derived branch infers team strength from spreads and totals. It has direct eligible quote pairs for 14 upcoming Week 7 games, only 7.82% of the 179 remaining fixtures. Later games are forecast from those team states. The branch is therefore not a sportsbook quote for the entire remaining schedule or for every season event.

Recorded quantity	Status at the forecast freeze
Data commit	20:35:14 UTC; 106 seconds before cutoff
Downloaded frozen bytes	20:40:53 UTC; later retrieval of the pinned earlier commit
Week 7 game quote pairs	14 admitted
Week 8 quote pairs	14 excluded
Historical / current-season quote rows	1,424 / 107; individual quote-update times unverified
Direct futures rows	160; all later comparators, none fitted

Denver-Arizona's pinned moneyline was Denver -238 / Arizona +195. Proportional two-way normalization gives Denver 67.5%; the ensemble gives 63.6% conditional on no tie, or 63.5% unconditionally. A spread of Denver -4.5 and total of 43.5 were recorded, but spread/total juice was unavailable. No -110 price is assumed.

What the audit can establish

The pinned snapshot existed before the cutoff. Individual quote ages and uniform bookmaker/closing definitions are not verified. An empirical stale-price effect or executable Tuesday-market backtest cannot be recovered from these records; arbitrary price shocks would not repair that missing provenance.

Method M3; evidence F6-F7 and S01-S02. NFL sources and model outputs were not refreshed with post-cutoff football information. A later download, later publication and later quote observation are distinct provenance facts.

PLAYERS / NAMED CONDITIONAL SCENARIOS

Player availability changes the path

Denver's clearest named scenario concerns Patrick Surtain II and Talanoa Hufanga. At the cutoff, both had recorded absences and no authenticated Week 7 clearance. Pat Bryant's completed injured-reserve placement establishes a minimum absence; Week 9 is his earliest modeled eligibility, not a predicted return. Riley Moss's Week 6 participation is retained. [S08-S09; F4]

Conditional assumption	Pool PO	PO change, pp	Pool SB
Denver defensive backs return in Week 7	69.7%	+0.31	4.24%
Denver defensive backs absent through Week 10	69.0%	-0.40	4.17%
DBs Week 7; Bryant Week 9	69.8%	+0.39	4.25%
Jackson and Mayfield absent through Week 10	69.4%	+0.03	4.22%
Hurts absent for the remainder of the 2026 season	69.3%	-0.04	4.21%
49 unknown-start absences stay absent	69.4%	+0.05	4.24%

Read the component effect and the pooled effect

Forcing the two defensive backs to return in Week 7 raises the availability branch's playoff chance by 2.06 percentage points. Keeping them absent through Week 10 lowers it by 2.70 points. The final ensemble changes by only +0.31 and -0.40 points because that branch carries 15.0% weight; the other five branches do not respond to the named-player intervention.

Method M4: replace only the availability-family event probability using each previously registered 500,000-world scenario: P_partial = P_primary + w_availability (P_scenario - P_availability). No scenario probability is assigned. Effects are conditional sensitivities, not clinical prognoses or causal player values. Tiny signed differences can be simulation noise; no direction is claimed for negligible effects. [F4]

PLAYERS / FOOTBALL INTERPRETATION

What to say about individual players

Surtain and Hufanga: monitor participation, not a promised date

The named early and delayed scenarios create a 4.75-percentage-point spread in the availability branch's playoff probability. Week 10 is Denver's bye; absence through that week forces misses in Weeks 7-9. Normal modeled recovery resumes in Week 11 without guaranteeing participation. A forced return also permits subsequent injury onsets. [F4]

Bryant: eligibility and readiness are different inputs

The earliest-eligibility scenario allows Bryant to participate in Week 9 alongside Week 7 returns for the two defensive backs. It produces a partial-pool playoff estimate of 69.8%, versus 69.4% primary. The model does not estimate his medical readiness, target share or receiving-yard distribution. [F4]

Bo Nix and the offense: a unit-level question

The positional diagnostic ranks Denver fourth in adjusted offensive sack avoidance, but twenty-third in net passing production and twenty-fifth in completion production. Those measures combine quarterback decisions, protection, receivers and scheme. They suggest evaluating the offense beyond sacks; they do not identify Nix's individual contribution or support a new passing-yard, touchdown or MVP probability. [F5]

Jackson, Mayfield and Hurts: league effects remain conditional

The package includes Jackson and Mayfield remaining absent through Week 10, and a separate Hurts season-absence scenario. Their opponents and playoff competitors change too. The primary Hurts input uses the authenticated IR minimum; expected season-long absence is kept as a separate scenario. No joint recovery probability is invented. [F4]

Method M4 / M5: named participation restrictions change one measured availability state per player. The model uses positional group coefficients rather than individualized replacement talent. No newly validated player counting-stat model was added to the completed ensemble. Player-specific conclusions here are participation sensitivities and clearly labeled unit diagnostics.

EXPERIMENTAL CHALLENGER / KEPT OUTSIDE THE PRIMARY POOL

The positional model earned scrutiny

A possession simulator models field position, clock, score, availability and measured matchup inputs. It includes passing production, completion, disruption, yards after catch, rushing, kicking, punting, play-calling tendencies and two interaction terms. An independently fitted reference uses the same engine without the eleven unit inputs. [F5]

Forecast branch	DEN playoffs	DEN title	Role
Same-engine reference	67.7%	2.74%	Experimental control
Unit matchup challenger	70.4%	3.18%	Experimental challenger
Primary ensemble	69.4%	4.21%	Headline forecast

2025 diagnostic	Reference	Unit challenger	Interpretation
Near-term win-credit Brier	0.22887	0.22789	Small improvement; paired interval spans zero
Multiweek win-credit Brier	0.25355	0.25605	Challenger worse; both above 0.25 benchmark

Research decision: no promotion

The near-term Brier difference is -0.00099 with a descriptive 95% week-cluster interval of [-0.00470, +0.00293]. The longer-horizon result deteriorates. Added detail has not demonstrated incremental predictive value, so neither positional branch receives ensemble weight.

Method M5: this is a unit-informed possession model, not a validated collection of individual QB/OL/DL/secondary/coaching agents. Proxy measurements do not isolate player ability. Rare clock and kick sequences remain incompletely validated. Earlier win-credit scores assign half credit to ties; do not directly compare their binary log losses with the later three-class ensemble table. [F5]

EXPERIMENTAL DIAGNOSTICS / NO CAUSAL PLAYER GRADES

Denver's measured matchup profile

Adjusted unit measure	Offense rank	Defense rank
Net passing yards per dropback	23	2
Completion fraction per nonsack attempt	25	1
Sack avoidance / sack generation	4	2
Yards after catch production / prevention	10	3
Designed rushing production / prevention	23	14

The most coherent strength in these inputs is pass defense: completion prevention, net passing prevention, sack generation and yards-after-catch prevention all rate highly. The offensive indicators are uneven. Strong sack avoidance does not automatically become efficient passing production. These are questions for film and scheme review, not instructions to change personnel.

Ranks use shrunken, opponent-adjusted point estimates with an eight-football-week half-life; older-season observations still contribute. There are no rank confidence intervals. The kicking residual ranks thirtieth on only 20.7 weighted attempts and is especially noisy. Coaching tendencies measure neutral passing and short-fourth-down aggression, with no causal coaching-quality ranking.

Why the detail remains diagnostic

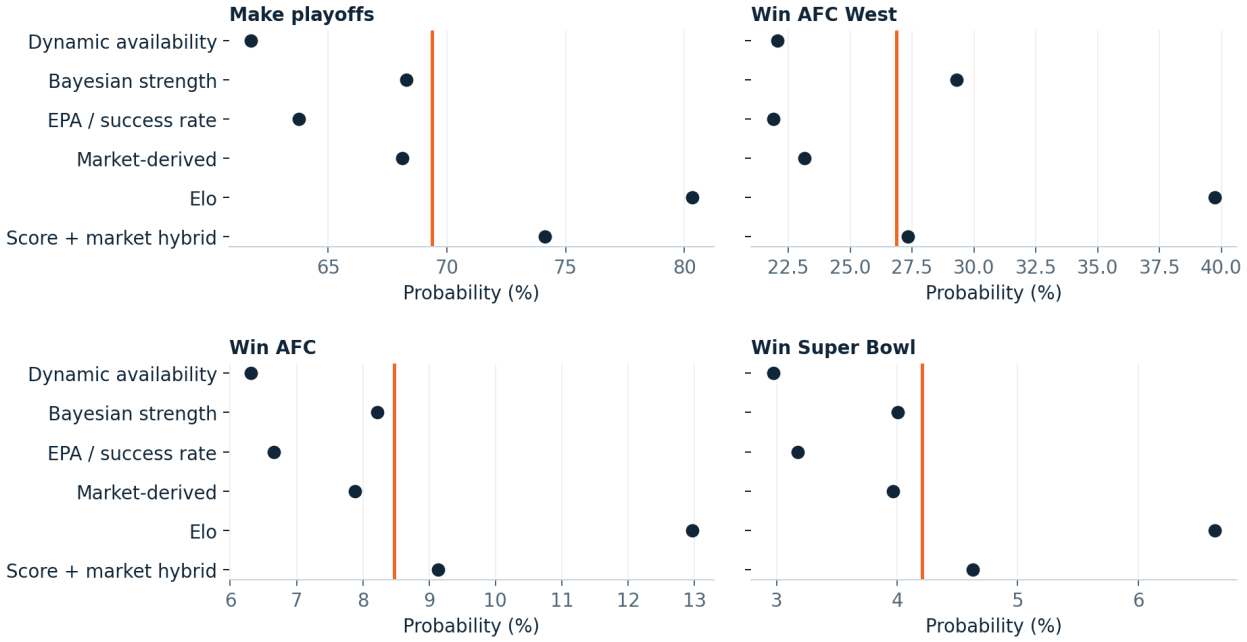
Only 3 of 88 marginal coefficient intervals exclude zero at an unadjusted 95% level. Correlated inputs, limited samples and multiple comparisons make selective coefficient stories unreliable. The matched final-game prediction test controls promotion, not how compelling an individual coefficient sounds.

Method M5; frozen unit and coefficient exports F5. Positive main-unit participation establishes observed participation, not complete health. A descriptive team rank cannot be assigned wholly to Bo Nix, an offensive lineman, a defensive back or a coach.

METHODS / MODEL MAP

Six admitted views of one season

Family	Weight	Main evidence / treatment
Score + market hybrid	18.1%	Results + score/market state
Elo	14.9%	Results / margin-adjusted rating updates
Market-derived	21.2%	Spread and total quotes / dynamic filter
EPA / success rate	16.7%	Opponent-adjusted EPA and success rate
Bayesian strength	14.1%	Unique game scores / full state-parameter posterior
Dynamic availability	15.0%	Joint score, measured absence and availability dynamics



Dots are complete component forecasts; orange lines are the fixed ensemble. Differences reflect assumptions and evidence, not simulation uncertainty alone. [F2]

The current primary mixture uses two million retained season paths per family: twelve million component worlds. Exact weighted counter pooling adds zero new ensemble worlds. Original v1, fixed-parameter, held-availability and positional controls receive no extra votes. The public efficiency family is not proprietary DVOA. [F2-F3]

METHODS / THREE PARALLEL EVIDENCE STREAMS

Elo, market and efficiency methods

Elo: results update a relative rating

A zero-sum Elo system updates after weekly results, using a logarithmic margin multiplier, an autocorrelation correction and offseason regression. The current native selection uses K = 20, a 40-point home rating adjustment and two-thirds offseason carry. A development-only point conversion maps rating differences to margins. Final probabilities come from the calibrated common score model, not directly from a logistic Elo formula. Rating covariance is approximate. [F2; S25]

Market-derived: quotes reveal a noisy team state

A dynamic Gaussian filter observes spreads and totals after removing home/rest context. Game results do not directly update this branch. Current native net filtering uses persistence 0.90, weekly variance 4 and quote SD 1 point. Total filtering uses persistence 0.97, quote SD 1 and weekly variance 0.20. These historical filter settings differ from separately fitted future trajectory parameters. Unknown later game quotes are inferred, not invented as posted lines. [F2, F6]

Efficiency: public EPA and success rates

Recency-weighted ridge models separate offensive and defensive effects for dropbacks and rushes, adjusting for opponents. Provider success means EPA greater than zero; pass flags include sacks and scrambles. A chronological ridge conversion maps prior-game features to margin and total. The current native grid selects an eight-playing-week half-life, 200-equivalent-play penalty, 0.25 offseason observation carry and both EPA and success features. [F2; S26-S27]

Shared limitations remain visible

The common score calibration uses fixed fitted parameters for these branches. Approximate state covariance omits some tuning and parameter uncertainty. Net/total state cross-covariance is omitted in this engine, even though game residual correlation is retained. Several fitted covariance scales hit the lower boundary 0.05; that is a diagnostic, not proof that true uncertainty is negligible.

These descriptions concern the current production states. Historical reconstruction refits eligible native and calibration parameters using only earlier years for each fold. Published mechanics are adapted rather than claimed as exact reproductions of another provider. [F2-F3]

METHODS / BAYESIAN TEAM STRENGTH

Updating and persistence do different jobs

Bayesian updating learns from observations already received. Persistence describes how strength can evolve before future observations arrive. The model needs both: a posterior over today's strength and parameters, plus a law for future change. Replacing persistence with updating alone would leave the future evolution unspecified.

Core state law

Net strength next week = rho x current net strength + centered Normal innovation.
Game mean margin = home strength - away strength + home-field and clipped rest context.
Game mean total = 45 + both teams' total-environment states.

Each completed regular-season game enters once as a margin/total pair. A Kalman filter integrates the 64-dimensional state conditional on parameters. Net states sum to zero; total states include a learnable league-wide mode. The learned full state covariance is retained, including posterior net/total dependence. Ten global parameters include home advantage, residual scales and correlation, persistence, weekly innovation scales and offseason transitions. [F3]

Four-chain MCMC integrates global-parameter uncertainty. Weekly forecasts update conditional states and parameter weights using only completed prior games; low importance support triggers a fixed-budget prior-prefix refit. Current forecast seasons draw one posterior component and one joint state, then carry that common trajectory through every game. Simulated future results do not become new real evidence or trigger an artificial refit.

The Stage 6 reconstruction retained 2.16 million MCMC draws across nine fits, with maximum reported R-hat 1.00350, minimum bulk effective sample size 2,924 and minimum tail effective sample size 1,729. The current Bayesian season bank uses 1,024 exported components. These diagnose numerical exploration of the specified posterior, not whether the football model is correct. [F3]

The Bayesian score likelihood is Gaussian, then mapped to rounded final scores and a statistical tie rule. Parameter integration does not guarantee heavy-tail calibration or exact NFL score support. Priors, initial states, rest coefficient and process assumptions remain conditional choices. [F3; S28-S30]

METHODS / MEASURED CONTRIBUTORS AND FUTURE PARTICIPATION

A joint availability model

The availability branch jointly fits score strength and six signed positional absence coefficients: QB, OL, offensive skill, DL, LB and DB. It does not deduct an arbitrary injury penalty from a market rating. Historical participation/exposure and score data determine the joint posterior, reducing mechanical double counting while leaving observational identification limits. [F4]

The frozen last-game panel contains 1,410 records. Of those, 1,148 contributors have eligible measured roles and known available/absent states for forecasting. An absence weight is based on up to three earlier healthy same-team games, with a 0.25 snap-share eligibility threshold. Weights freeze at absence onset; injury zeros do not erase established roles. Specialists, unknown states and unmeasured roles are not silently treated as healthy starters.

Position-specific onset and return rates are estimated from 2021-24. Return hazards depend on completed missed team games in age bands 1, 2, 3, 4-6 and 7+. A player has at most one state transition before a scheduled game. Byes do not age a game-count spell. Participation is the outcome being modeled, not medical clearance or restored effectiveness.

Rate uncertainty uses 4,096 shared Bayesian cluster-bootstrap configurations over 128 team-seasons. One configuration governs all positional cells within a world. Smoothed event rates use (weighted events + 0.5)/(weighted opportunities + 1). The full season also shares a joint score/availability-coefficient posterior draw. Original primary random transitions are conditionally independent across players given those common parameters; new stress tests add shared weekly adversity.

News restrictions are explicit

Authenticated IR minima and club-confirmed season absences impose hard restrictions only on uniquely mapped eligible contributors. Restrictions never rewrite historical exposure. Overlapping scenario/news effects create one absence state. Eligibility allows modeled recovery; it does not force a return. Incomplete news coverage and 49 unknown-start current spells remain limitations.

Method M4; original contracts, cohort, news ledger and compressed posterior authority: F4. The coefficient model estimates group-level associations and cannot resolve individual replacement quality or diagnosis-specific recovery. [S08-S09; S31]

METHODS / GAME SIMULATION AND SEEDING

From team states to a complete NFL season

Each component fixes 93 completed scores and simulates the 179 remaining regular-season fixtures. It then applies division/wildcard ranking, creates a seven-team field per conference, awards the top-seed bye, reseeds the divisional round and simulates thirteen postseason games through one Super Bowl winner. All 32 teams inhabit the same world. [F1-F3; S10-S12]

The hybrid, Elo, market and efficiency families use correlated Student-t margin/total residuals with a shared tail shock. Seven score/trajectory parameters are fitted separately for each family, including residual scales, tail degrees of freedom, residual correlation, future persistence, innovation variance and origin covariance scale. Team states persist across games; a schedule is not a collection of independent coin flips.

Bayesian and availability families use their joint Gaussian posterior score law, with uncertainty over parameters and states. Both score engines convert margin/total draws into nonnegative rounded points. A calibrated statistical rule resolves most rounded ties; regular-season ties remain possible. Postseason ties are resolved to a winner. The Super Bowl has zero modeled home-field term, even though a nominal home designation exists.

Important score-model boundary

This is a distributional season forecast. The statistical score support admits impossible NFL pairs such as 1-0 and 1-1; it does not enumerate touchdowns, field goals, safeties or possession overtime. That approximation limits exact-score, key-number, spread/total and overtime-process interpretations. The positional challenger explores a different engine but was not promoted.

Engine audit: unchanged seeding evidence includes fifteen focused tie cases and seventy exact historical seed placements. New composed wildcard, tie-heavy schedule and label-permutation checks passed. A net-touchdown tie-break cannot be reconstructed from aggregate score draws; the rare fallback is recorded and bounded separately. [F8; S10-S12]

METHODS / CHRONOLOGY AND COHERENT POOLING

How the ensemble weights were learned

Regularized predictive mixture

Choose nonnegative weights that sum to one, minimizing:
weighted three-class log loss + lambda x sum of (weight - equal-family weight)^2.
The selected lambda is 0.1. Weights combine predictions; they are not probabilities that the models are true.

For reconstructed 2023 forecasts, native/calibration selection uses 2022 with 2021 initialization. For 2024, it uses 2022-23. Bayesian likelihoods and availability transitions stop before each forecast year. Current-season weekly states may update with games already completed; they never see the target game result. This repairs parameter chronology without claiming contemporaneous archived data vintages. [F3]

Half the ensemble objective belongs to near-term forecasts and half to multiweek forecasts. Years receive equal mass within a scope; multiweek origins 4, 7, 10 and 14 receive equal mass within year. A five-value lambda grid is fitted on 2023 and selected on 2024 using the declared paired one-standard-error rule. All five qualify, so the strongest shrinkage, 0.1, is selected. Final weights refit on 2023-24.

The completed historical panel has 1,629 origin-target rows but only 512 distinct games. The 2025 diagnostic has 815 rows representing 256 distinct games. Week 1 is excluded for a common availability cohort. Repeated multiweek targets are kept together in target-week cluster resampling. No claim treats all rows as independent observations.

Whole-season mixture: P(event) = sum of weight x component event frequency. Model identity is chosen once for an entire season, preserving each family's joint strength, availability, seeding and playoff structure. Conditional probabilities pool their joint numerator and denominator first. Exact pooling reuses counters instead of simulating another arbitrary mixture sample.

Method M6; frozen protocol and weights: F3. Study-wide design decisions and earlier 2025 inspection remain exposed. The stacking and distribution-combination literature motivates predictive pooling; it does not validate this particular NFL model or automatically establish calibration. [S28-S30]

VALIDATION / EXPOSED DIAGNOSTICS

What the historical evidence supports

Forecast	Near-term log loss	Multiweek log loss
Ensemble	0.65157	0.68251
Market-derived	0.63757	0.68327
Score + market hybrid	0.65219	0.68530
Elo	0.66620	0.68721
EPA / success rate	0.66814	0.69510
Bayesian strength	0.66346	0.68202
Dynamic availability	0.65543	0.68060
Equal Family	0.65300	0.68256

Lower log loss is better. Every row uses home win, tie and away win as three separate outcomes. The ensemble's near-term loss is 0.65157 versus 0.63757 for the market branch. The market advantage is visible in descriptive paired target-week intervals. Quote timing is unresolved, so the streams do not certify identical historical information clocks or a causal method advantage.

At the multiweek horizon the ensemble is 0.68251 and the market branch 0.68327. Their difference is small, and its descriptive interval spans zero. Availability and Bayesian components also compete closely at this horizon. No exposed diagnostic justifies selecting a new winner after seeing these scores.

What is not a fresh holdout

2024 selected the ensemble penalty; it is tuning evidence. 2025 was already seen in earlier research; it is exposed diagnostic evidence. Historical public data have revisions and incomplete individual quote timestamps. There is no untouched test season in this retained evidence set.

Method M6: scoring probabilities are floored at 10^{-6} per class and renormalized for scoring only. This does not modify stored season probabilities. Binary ensemble diagnostics count actual home wins, with ties as non-home-wins; earlier half-credit tables use a different target. No ensemble CRPS is manufactured by averaging component CRPS values. [F3, F7]

VALIDATION / FOUR KINDS OF UNCERTAINTY

Precision is not the same as certainty

Source of uncertainty	What this package shows	What it does not establish
Finite simulation	DEN title MC standard-error upper bound: 0.014 percentage points	Accuracy of the football assumptions
Weights from limited history	Fixed-lambda cluster range for DEN title: 4.12-4.31%	A total forecast confidence interval
Choice of model family	Six component title estimates: 2.97-6.63%	A calibrated probability interval
Future outcomes / structure	Win distributions and registered stress cases	Complete coverage of future shocks or model failures

A high simulation count makes the specified model's answers numerically stable. It cannot create more observed seasons, repair missing injury information or certify the probability of a rare championship. The narrow weight-bootstrap range is conditional on the fixed candidate set, penalty and forecast banks.

Six-family forecast correlations are high: approximately 0.864-0.974 near term and 0.879-0.984 multiweek. Shared outcomes also make residual correlations high. These numbers do not measure independent information. Controls and copied models receive no extra votes; a duplicate-label accounting check leaves the mixture unchanged to floating-point precision.

Descriptive home-win calibration slopes are about 1.10 near term and 0.69 multiweek for the ensemble, with broad intervals. The lower multiweek point estimate suggests checking excess probability spread, but the interval includes one. Earlier component multiweek margin intervals also undercovered nominal 80% levels. Game-level diagnostics do not independently validate division or title-event calibration.

Methods M6-M7; F2-F3 and F7. Weight sensitivity uses 1,000 target-week cluster refits with lambda fixed; family exclusion ranges are diagnostics of the chosen perturbations. Simulation MC error is conditional on retained posterior banks and weights, separate from MCMC integration and structural uncertainty.

ADVERSARIAL TESTING / REGISTERED FAMILY REMOVAL

Stress test: remove shared market inputs

Case	DEN PO	DEN division	DEN SB
Frozen ensemble	69.37%	26.87%	4.208%
Market family alone	68.12%	23.14%	3.965%
Remove both; fixed renormalization	68.38%	28.04%	4.168%
Remove both; refit survivors	68.48%	28.15%	4.188%

The market and hybrid families jointly carry 39.33% of the frozen weight. Both are removed because both directly use quote evidence. One case simply renormalizes the surviving weights; the other refits the four survivors on 2023-24 at the unchanged lambda 0.1. No new season worlds are generated.

Denver is stable; the league is not uniformly stable

Denver's refitted four-family title estimate is 4.188%, only 0.0206 percentage points below primary. The conservative conditional MC error bound for that difference is 0.0111 points. Dallas's playoff probability falls 11.21 points, the Giants' rises 7.86 and San Francisco's title chance rises 1.39 points. Market exposure can matter greatly even when Denver's headline barely moves.

On exposed 2025 near-term games, excluding both and refitting worsens log loss by 0.00891, with a descriptive paired interval [0.00389, 0.01373]. Its multiweek difference from primary is unresolved. Removing market families is therefore a robustness probe, not evidence that the prices were useless.

Method M7: four predeclared cases, unchanged loss design, one four-family refit, and 2,000 paired target-week cluster draws. A chronological 2023-only fit supports a separately labeled 2024 tuning diagnostic. Surviving families may retain inherited calibration or research lineage from market evidence; this is not causal removal of all historical market influence. [F7]

ADVERSARIAL TESTING / PERSISTENCE AND INNOVATION

Stress test: how team strength evolves

Registered case	Pool PO	PO change, pp	Pool SB	SB change ± MCSE, pp
Half strength-innovation variance	69.42%	+0.06	4.202%	-0.006 ± 0.003
Double strength-innovation variance	69.26%	-0.10	4.218%	+0.010 ± 0.003
Persistence fixed at 0.90	69.12%	-0.25	4.251%	+0.042 ± 0.004
Persistence fixed at 1.00	69.40%	+0.04	4.194%	-0.014 ± 0.003

All four tests keep the current Bayesian posterior and Week 7 state fixed. They change only future net-strength innovation variance or persistence. Multiplying variance by one-half or two changes innovation standard deviation by its square root; it does not halve or double the starting team rating. Total-score evolution, fitted coefficients, residual score law, schedule and seeding stay fixed.

What the largest tested Denver change means

Persistence fixed at 0.90 produces the largest absolute title change among these four tests: +0.042 percentage points in the partial ensemble, to 4.251%. The corresponding Bayesian-branch change is +0.301 points. Only the 14.10%-weight Bayesian family changes; this is not an all-model perturbation.

Each stress uses 500,000 matched worlds against a 500,000-world control. Posterior component selection, starting states, innovations and game-slot shocks share purpose-separated random streams. When brackets change, playoff shocks remain aligned by round slot. This pairing reduces noise in changes without assuming opponents remain identical.

Method M7: partial P = frozen primary P + branch weight x (matched stress P - matched control P). MCSE comes from actual per-world paired differences. The displayed ± number is one Monte Carlo standard error, not total forecast uncertainty or a probability for the scenario. No stress was selected to replace the headline. [F9]

ADVERSARIAL TESTING / CORRELATED PARTICIPATION PATHS

Stress test: absences arrive together

Registered case	Branch PO	Branch change, pp	Pool PO	Pool SB
Within-team latent correlation 0.50	61.48%	-0.20	69.34%	4.220%
Within 0.50; cross-team 0.20	61.57%	-0.10	69.35%	4.219%
Clustered adversity + delayed DEN DBs	58.92%	-2.76	68.95%	4.180%

The original transition model already shares uncertainty in fitted onset and return rates. These tests add weekly shared adversity across individual transitions. A Gaussian construction sets total within-team latent correlation to 0.50; the cross-team version sets common league correlation to 0.20. In the latter, variance shares are 0.20 global, 0.30 team and 0.50 individual.

Each conditional one-step transition probability is preserved. The common factor's direction is reversed for return versus onset, so adversity raises joint unavailability rather than generating simultaneous onset and recovery. Fresh global and team factors arrive each week. These are latent correlations, not measured binary injury correlations or estimated medical contagion rates.

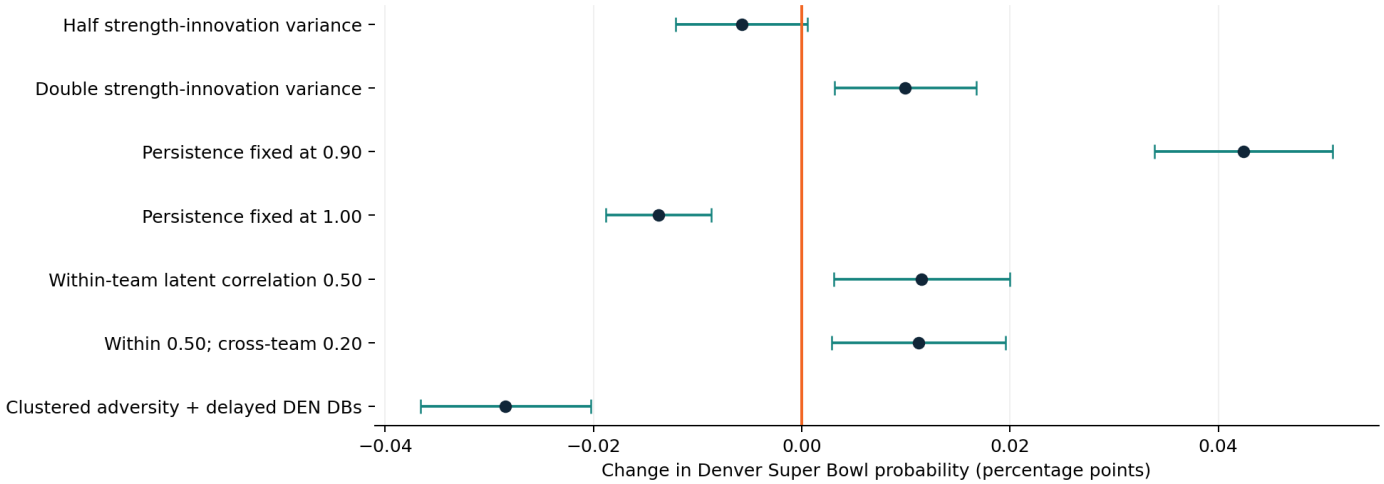
The registered adverse Denver path

Combining clustered adversity with the existing Surtain/Hufanga absence-through-Week-10 assumption yields 68.95% partial-pool playoffs and 4.180% title probability. The title change is -0.028 points, with paired MCSE 0.004 points. This scenario changes participation states once; it adds no duplicate rating penalty.

Method M7: three 500,000-world stresses share a 500,000-world availability control. Only the 15.00%-weight availability branch responds. No clinical probability or all-ensemble causal effect is assigned. Bounds, event nesting, all-team conservation and matched sufficient statistics are checked. No case combines the Bayesian and availability stresses. [F9]

ADVERSARIAL TESTING / DENVER AND THE LEAGUE

Where the registered stresses matter



Denver: changes in partial-ensemble title probability. Error bars are ±1.96 paired Monte Carlo standard errors of the change only; they exclude the primary point's simulation error and structural uncertainty. [F9]

Largest absolute title shifts across registered team/scenario pairs

Team	Registered case	Title change, pp
JAX	Persistence fixed at 0.90	-0.361
SEA	Persistence fixed at 0.90	-0.236
NYG	Persistence fixed at 0.90	+0.157
SF	Persistence fixed at 0.90	-0.151
CLE	Persistence fixed at 0.90	+0.148

These are marginal sensitivities of one changed branch at a time. A stable Denver result would not establish stability for every team, and the largest changes are not probability-weighted downside forecasts. The full all-team table retains every case rather than only these selected extremes.

Method M7: display the five largest absolute title-probability changes among the seven registered stresses and all 32 teams. No outcome changes the frozen weights or selects a replacement model. Different stresses are not combined into a single forecast interval. [F9]

ADVERSARIAL TESTING / IMPLEMENTATION REVIEW

Stress test: the engine and its limits

Test or audit	Result / scope
Actual family source bindings	Shared seeding and appropriate score laws verified by hashes
Composed wildcard re-entry	Passed; lower division club re-enters after representative qualifies
32 tie-heavy schedule completions	Passed: 17 games/team, standings conservation and unique seeds
Team-index permutation	Same seeds in all 32 resolved synthetic fields
Actual score-function edge probes	32 calls total; rounding/clipping/tie behavior as documented
Retained previous evidence	15 focused tie tests and 70 historical seed placements reused

No blocking hard defect was established for the registered frozen inputs. The audit does not certify every possible tiebreak state or arbitrary numerical input. The original score/seed implementation remains unchanged, and the primary weights remain frozen. [F8]

Football realism remains a material boundary

Both statistical engines admit score pairs such as 1-0 and 1-1. At a fixed expected margin of zero and total of 45, an analytic residual-law table quantifies clipping and rare tails; it is not an integrated real-football tail forecast. The postseason tie mechanism determines a winner without simulating actual overtime possessions.

One Elo ranking decision reached the unavailable-net-touchdowns fallback; the other five retained branches recorded none. At most 7.45505×10^{-8} of empirical mixture probability mass reached that fallback. The counter can include lower division rankings, so this is not an observed playoff-impact rate or a population confidence bound.

Method M7; F8. Input-domain guards and checked 32-bit score conversion are not comprehensive; no registered-input violation was found. The positional engine's rare kickoff-template issue remains disclosed on its zero-weight branch. Selected mechanical passes do not establish exact score support, causal effects or championship calibration.

METHOD INDEX / HOW TO AUDIT THE CONCLUSIONS

Every prediction has a defined method

Prediction	Method and evidence	Main interpretive boundary
Final wins and standings	M1: full season counters; F1-F3	Means and marginal modes are not one jointly predicted table
Playoff / division / AFC / title	M1: complete seeding and postseason in each world	Rare-event calibration not independently established
Playoff and Super Bowl opponents	M2: pooled joint event counters divided by stage probability	Conditional on qualification and simulated survival
Market value comparisons	M3: observed purchase price, stated payout math; F6-F7	Later snapshots, unverified costs and execution
Named-player availability cases	M4: explicit restrictions within the fitted availability branch	Unweighted scenarios; partial-pool, not causal player effects
Unit and coaching findings	M5: shrunken matchup proxies and same-engine comparison; F5	Experimental; no individualized ability or causal coaching grade
Ensemble and weight sensitivity	M6: chronological prediction panels, regularized pooling; F3	Retrospective/exposed diagnostics and finite data
Adversarial scenarios	M7: registered removals, dynamics and dependence changes; F7-F9	Selected stress cases, not probability-weighted alternative futures

Percentages are rounded for readability; CSV files retain full precision. Each scenario table states its denominator. A paired stress standard error measures finite simulation error in that particular difference; it is not a confidence interval for a clinically or causally identified effect.

SYNTHESIS / MATERIAL LIMITS AND NEXT EVIDENCE

What would change the forecast?

New evidence updates the forecast; a simulation does not

New completed games, authenticated participation changes and fresh properly timestamped prices can change the posterior state and forecast. This edition is a frozen October 20 analysis, not a live dashboard. The model does not know future results merely because millions of paths have been generated.

The headline is a model-conditional probability

Denver's 4.2% title chance is the current six-family mixture result. It is compatible with meaningful model disagreement, missing information and imperfect cross-game dependence. It should not be advertised as accurate to two decimal places in the real world merely because simulation noise is small.

Availability and matchup detail have limits

The injury model measures participation among a frozen cohort, with incomplete news, unknown starts, role exclusions and group-level coefficients. The positional branch does not isolate individual ability and did not earn promotion. Scenarios are useful for conditional planning; they do not establish return dates or the causal benefit of a tactical change.

Prospective evaluation remains the decisive next test

The package repairs chronological fitting, retains exposed diagnostics and tests specific assumptions. It does not possess an untouched historical test season or calibrated championship-frequency evidence. The next meaningful validation is an archived prospective forecast, evaluated against subsequent results with the scoring definitions fixed in advance.

The completed research judgment

Retain the six-family forecast and its explicit uncertainty. Publish the weaker historical results, the nonpromoted positional model and the bounded stress findings alongside the attractive headline. The work is more credible because the limits are visible; additional simulation count alone would not resolve them.

Evidence F1-F9. No optional stress result is used to retune the headline weights. Any future forecast should receive a new dated information cutoff and retain this edition for prospective comparison.

EVIDENCE REGISTER / EXTERNAL AUTHORITIES

Sources and provenance

S01 / Pinned games, lines, teams and standings
nflverse/nfldata commit 603d490d, created before cutoff; exact downloaded bytes and schemas retained.
Open primary source

S02 / nflverse field definitions
Spread signs and source schema; individual quote times are not certified.
Open primary source

S08 / Denver injury reports and dated club updates
Use frozen dated captures in F4, not a refreshed live status page.
Open primary source

S09 / Broncos Bryant IR transaction
Authenticated placement supports the modeled minimum, not a return promise.
Open primary source

S10 / 2026 official NFL playing rules
Official scoring and overtime reference; the primary score generator is a disclosed approximation.
Open primary source

S11 / 2026 NFL Record and Fact Book
Frozen official excerpts support seeding/restarters and neutral-site interpretation.
Open primary source

S12 / NFL tiebreaking procedures
Official standings reference; source text and rule tests are preserved.
Open primary source

S13 / Kalshi public futures API
All later captures and event-specific URLs retained in F6; descriptive comparison only.
Open primary source

EVIDENCE REGISTER / EXTERNAL AUTHORITIES

Sources and provenance / continued

S15 / BetMGM all-team Super Bowl market
Later frozen +2000 Denver quote; proportional no-vig normalization is a stated convention.
Open primary source

S25 / Published Elo mechanics
Mechanics reference; the fitted SSFA model is an adaptation.
Open primary source

S26 / nflreadr play-by-play loading and updates
Pinned release asset metadata and provider revisions retained in Stage 2 sources.
Open primary source

S27 / nflfastR cleaned-play definitions
EPA/success and play-type definitions; not proprietary DVOA.
Open primary source

S28 / Yao et al.: predictive stacking
Methodological motivation for predictive combinations; no NFL performance claim.
Open primary source

S29 / Burkner, Gabry, Vehtari: leave-future-out validation
Chronological predictive evaluation reference; this study also discloses design exposure.
Open primary source

S30 / Gneiting and Ranjan: combining distributions
Pooling, dependence and calibration considerations.
Open primary source

S31 / Rubin: Bayesian bootstrap
Methodological basis for shared cluster-weight draws, adapted here to team-seasons.
Open primary source

EVIDENCE REGISTER / COMPUTED RESULTS

The calculations behind the report

F1 / Final primary probabilities and opponent joints
summary.json; league_projection.csv; remaining_games.csv; Denver record, seed, win and opponent tables.

F2 / Component forecasts and native models
Six two-million-world branches and original-stage dependency manifests.

F3 / Chronology, weights and historical diagnostics
preregistered_protocol.json; learned_weights.json; prediction panels; metrics; weight sensitivity and audits.

F4 / Availability contracts and named scenarios
Frozen news ledger, measured cohort, posterior authority, transition rates, scenarios; Stage 6 partial-pool translation.

F5 / Positional challenger and unit diagnostics
Matched game diagnostics, current unit exports, model contracts and source register; zero primary weight.

F6 / Frozen direct futures and game quote context
Full all-team comparator CSV; Denver observed prices and capture times; no later-market fitting.

F7 / Market/evidence stress review
Four-case outputs, source manifest, chronological/exposed diagnostics and market timing audit.

F8 / Score and seeding engine review
Source bindings, deterministic edge fixtures, tail and fallback scope, official-source evidence.

F9 / Matched dynamics and availability stresses
Restoration/implementation receipts, nine runs, paired sufficient statistics and conservative interpretation.

The external sources provide inputs and methodological context. Forecast probabilities, stress changes and validation scores are SSFA calculations, not forecasts published by those source providers. Exact relative paths, source digests and delivery manifests are in the accompanying reproducibility archive.

REPRODUCIBILITY / FINAL HANDOFF

Reproduce, inspect and extend

Start with the readable outputs

The PDF and offline HTML contain the same substantive report. The archive includes full-precision all-team, fixture, opponent, market and stress tables; the report specification and chart/render code; protocols; source bindings; verification receipts; and retained numerical reconstruction inputs. Consult the archive README for paths and dependency scope.

Reusing paths does not create new observations

The primary forecast pools twelve million component worlds: ten million reused from accepted earlier branches and two million generated for the restored Bayesian branch. Exact pooling generates no new worlds. The new adversarial dynamics work adds nine runs of 500,000 season evaluations, totaling 4.5 million. These runs reuse 500,000 matched underlying worlds per family; the 4.5 million evaluations are not independent observations. Market removal reuses counters and adds none. Prior scenario and challenger work is separately identified rather than double counted.

Restoration is explicit

Stage 6 preserves its numerical posterior reconstruction and compact banks. Stage 7 restores the original availability posterior array authority and rebuilds derived factors within its own subtree. It does not silently overwrite prior-stage files. Reproducing the forecast from retained processed inputs is distinct from rebuilding every public raw-data extraction or recovering historical publication vintages.

What a future extension should retain

Keep the data cutoff, source version, model eligibility, seed manifest, scoring target, denominator definitions and output hashes. Log new evidence separately. Save a prospective forecast before later results arrive, then evaluate it without changing its original weights. Expand structural modeling only when a defined diagnostic warrants it.

Edition identity

SSFA_Broncos_and_the_NFL_Complete_Forecast_20261020_v2
A new filename separates this consolidated report from the original PDF and intermediate HTML checkpoints. The seven executive summaries describe this entire completed edition.

Final build and delivery manifests bind the PDF, HTML and archive bytes. Numerical standard errors, posterior diagnostics and substantive limitations remain part of the deliverable rather than being removed for presentation.